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# Summer Exercises

## 1.1 Continuous probability space

### Exercise 1.1.1 ► Bertrand's Paradox

Show that the probability of the event: a randomly chosen chord of the unit circle passing at a distance of  $\frac{1}{2}$  from the center, is  $\frac{1}{3}$  or  $\frac{1}{2}$  depending on the chosen model. Compare the two models.

- Proof.* 1. Let  $\Omega$  be the collection of the pair  $(l, d)$ , where  $l \in [0, 2\pi]$ ,  $d \in [0, 1]$ . Let  $\mathcal{F} = \mathcal{P}(\Omega)$ . By randomly choosing a chord of the unit circle, we mean randomly choosing a  $(l, d) \in [0, 2\pi] \times [0, 1]$ , then we identify the  $(l, d)$  with the chord whose radius is  $l$  and the distance between the center point of the chord with that of circle is  $d$ . Let  $P$  be the normalized Lebesgue measure on  $[0, 2\pi] \times [0, 1]$ , then the probability we discuss is  $\frac{1}{2}$ .
2. Let  $\Omega'$  be all the points on the boundary of the unit circle. Let  $\mathcal{F}' = \mathcal{P}(\Omega')$ . First, fix a point  $p$  on  $\Omega'$ . By choosing a chord of the unit circle, we mean randomly choosing a point  $q \in \Omega'$ , and identify the point with the chord connecting  $p$  and  $q$ . Since the chord is  $\frac{1}{2}$  passing at the center if and only if the angle (which is range from 0 to  $\pi$ ) corresponding to the chord is greater than  $\frac{\pi}{3}$ . Similarly let  $P$  be the normalized Lebesgue measure on  $(\Omega', \mathcal{F}')$ . Then the probability we discuss is  $\frac{1}{3}$ .

□

### Exercise 1.1.2 ► Change of Variables

1. Let  $X$  be a uniform random variable on  $]-\frac{\pi}{2}, \frac{\pi}{2}[$ . What is the distribution of  $\tan X$ ?
2. Let  $X$  be a random variable with the density  $f(x) = \frac{1_{[0,1]}(x)}{\log 2(1+x)}$  show

that  $\frac{1}{X} - \lfloor \frac{1}{X} \rfloor$  has the same distribution as  $X$ .

*Proof.* 1. Denote  $\tan X$  by  $Y$ . Then

$$F_Y(x) = P(\tan X \leq x) = P(X \leq \arctan x)$$

$$F_Y(x) = \int_{-\infty}^{\arctan x} \frac{1}{\pi} 1_{\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)} dx = \frac{1}{\pi} \arctan x + \frac{1}{2}$$

the density function of  $F_Y$  is

$$f_Y(x) = \frac{1}{\pi} \frac{1}{x^2 + 1}$$

2. When  $x > 0$ ,

$$P\left(\frac{1}{X} \leq x\right) = P\left(X \geq \frac{1}{x}\right) = \int_{-\infty}^{\frac{1}{x}} 1_{[0,1]} \frac{1}{\log 2 (1+y)} dy =$$

□

### Exercise 1.1.3

Determine the expectation and variance of a normal distribution, gamma distribution, and log-normal distribution

*Proof.* 1. If  $X \sim \mathcal{N}(\mu, \sigma^2)$ , then the PDF is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Let  $y = \frac{x-\mu}{\sigma}$ , then  $x = \sigma y + \mu$ ,

$$\begin{aligned} E[X] &= \frac{1}{\sqrt{2\pi\sigma^2}} \sigma \int_{\mathbb{R}} (\sigma y + \mu) \exp\left(-\frac{1}{2}y^2\right) dy \\ &= \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-\frac{1}{2}y^2\right) d\left(\frac{1}{2}y^2\right) + \frac{\mu}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-\frac{1}{2}y^2\right) dy \\ &= 0 + \mu = \mu \end{aligned}$$

To calculate  $\text{Var}(X)$ , we assume that  $\mu = 0$ . Then

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] = E[X^2] \\ &= E[X^2] = \int_S (X(x))^2 dP \\ &= \int_{\mathbb{R}} x^2 dP_X \\ &= \int_{\mathbb{R}} x^2 f(x) dx = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx \end{aligned}$$

Let  $z = \frac{x}{\sigma}$ , then

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{\infty} \sigma^2 z^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}z^2\right) \sigma dz \\ &= \sigma^2 \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz \\ &= \sigma^2 \text{Var}(Z \sim \mathcal{N}(0, 1)) \\ &= \sigma^2 \end{aligned}$$

Specifically, consider

$$I(a) = \int_{-\infty}^{\infty} \exp\left(-a \frac{z^2}{2}\right) dz = \sqrt{\frac{2\pi}{a}}$$

and take differentiation of  $a$ .

2. A Gamma distribution has the density function

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \alpha, \beta > 0$$

*Note.*

事件以平均恒定的速率  $\beta$  发生, 等待第一次时间发生的事件服从指数分布, 等待第  $\alpha$  次时间发生的时间服从  $\Gamma(\alpha, \beta)$  分布.

$$E[X] = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$$

3. by a log-normal distribution  $X$ , we mean  $Y = \log X$  is a normal distribution. Then

$$F_X(x) = P(X \leq x) = P(Y \leq \log x) = F_Y(\log x), \quad x > 0$$

$$f_X = \frac{dF_X}{dx} = f_Y(\log x) \frac{1}{x} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right) \frac{1}{x}, \quad x > 0$$

□

#### Exercise 1.1.4 ► Couples of Random Variables

1. Let  $X$  and  $Y$  be independent random variables with distribution  $\mathcal{N}(0, 1)$ . Determine the distribution of the pair  $(X/Y, Y)$ , then that of  $Y$ . Are  $X/Y$  and  $Y$  independent?

*Proof.*

$$f_X = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad f_Y = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)$$

Let  $U = X/Y, V = Y$ , then  $X = UV, Y = V$ ,

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v)) \left| \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \right|$$

where

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) = \frac{1}{2\pi} \exp\left(-\frac{x^2 + y^2}{2}\right)$$

Then

$$f_{U,V}(u, v) = \frac{1}{2\pi} \exp\left(-\frac{u^2 v^2 + v^2}{2}\right) \left| \det \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix} \right| = \frac{1}{2\pi} \exp\left(-\frac{v^2(u^2 + 1)}{2}\right) |v|$$

$$f_Y(y) = f_V(v) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{v^2}{2}\right)$$

$$(f_{U,V}/f_V)(u, v) = \frac{|v|}{\sqrt{2\pi}} \exp\left(-\frac{u^2 v^2}{2}\right)$$

is not independent with  $u$ . Thus

$$f_{U,V} \neq f_U f_V$$

$\frac{X}{Y}$  and  $Y$  are not independent. □

### Exercise 1.1.5 ▶ Estimation of a Support Paramete

Let  $(X_n)$  be a sequence of i.i.d. random variables uniformly distributed on  $[0, \theta]$  ( $\theta > 0$ ) and  $M_k = \sup_{1 \leq n \leq k} X_n$ .

1. Calculate the density of  $M_k$ .
2. Calculate the expectation and variance of  $M_k$ .
3. We propose to estimate the parameter  $\theta$ . What do you think of  $M_k$  as an estimator, is it unbiased? Propose an unbiased estimator  $\hat{\theta}_k$  constructed from  $M_k$ . Show that this estimator converges in probability towards  $\theta$  and compare it to the unbiased estimator constructed from the empirical mean of the observations.
4. Show that  $k \left(1 - \frac{M_k}{\theta}\right)$  converges in distribution.

*Proof.* 1.

$$M_k \leq m \iff \forall n, X_n \leq m$$

$$P(M_k \leq m) = \prod_{n=1}^k P(X_n \leq m)$$

$$F_{M_k} = (F_X)^n = \left(1_{[0, \theta]} \frac{x}{\theta}\right)^n = 1_{[0, \theta]} \frac{x^n}{\theta^n}$$

$$f_{M_k} = \frac{dF_{M_k}}{dx} = 1_{[0, \theta]} k \frac{x^{k-1}}{\theta^k}$$

2.

$$E[M_k] = \int_{\mathbb{R}} x f_{M_k}(x) dx = \int_0^\theta k \frac{x^k}{\theta^k} dx = \frac{k}{\theta^k} \frac{1}{k+1} \theta^{k+1} = \frac{k\theta}{k+1}$$

$$\begin{aligned} \text{Var}(M_k) &= \int_{\mathbb{R}} x^2 f_{M_k}(x) dx - \mu^2 = \int_0^\theta k \frac{x^{k+1}}{\theta^k} dx - \mu^2 \\ &= \frac{k\theta^2}{k+2} \\ &= \theta^2 \left( \frac{k}{k+2} - \frac{k^2}{(k+1)^2} \right) \\ &= \theta^2 \left( \frac{k(k+1)^2 - k^2(k+2)}{(k+1)^2(k+2)} \right) \\ &= \theta^2 \frac{k}{(k+1)^2(k+2)} \end{aligned}$$

3.

$$E[M_k] - \theta \neq 0$$

it is not unbiased. Let

$$\hat{M}_k = \frac{k+1}{k} M_k$$

Then

$$\hat{\theta}_k = \theta$$

By chebyshev's inequality

$$a^2 E(|X| > a) \leq EX^2$$

only need to show that

$$\lim_{k \rightarrow \infty} \text{Var}(\hat{M}_k)$$

□